Triple-Higgs-boson probes of the Higgs potential and top–Higgs-boson interactions

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Abstract

We study triple-Higgs-boson production in the κ-framework as a direct probe of the Higgs potential. We derive complementary compact parametrisations of the inclusive gluon-fusion cross section: one retains the dependence on κ_3, κ_4, κ_t, and κ_{2t} for $\kappa_{3t} = 0$, while the other incorporates κ_{3t} for $\kappa_t = 1$ and $\kappa_{2t} = 0$. These results provide practical tools for interpreting current and future LHC measurements. We use the parametrisations to assess the sensitivity of LHC Run 2 data and HL-LHC projections to the Higgs boson self-couplings and show that differential kinematic distributions can provide additional discriminatory power between scenarios with nearly identical inclusive production rates.

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1 Introduction

The discovery of the Higgs boson in 2012 [1, 2] completed the particle content of the Standard Model (SM) and established the mechanism of electroweak (EW) symmetry breaking. Since then, a major objective of the LHC physics programme has been to determine the properties of the Higgs boson with increasing precision. While its couplings to EW gauge bosons and third-generation fermions are already constrained at the few- to ten-percent level, its self-interactions remain largely unexplored. Measuring the Higgs boson self-couplings is therefore one of the most important outstanding goals of collider physics, since they directly probe the shape of the Higgs potential and the dynamics of EW symmetry breaking.

In the SM, the Higgs potential is determined by a single quartic coupling $\lambda = m_h^2/(2v^2)$. After EW symmetry breaking, it predicts unique values for the Higgs boson trilinear and quartic self-couplings,

$$\lambda_3^{\rm SM} = \lambda\,, \qquad \lambda_4^{\rm SM} = \frac{\lambda}{4}\,, \tag{1}$$

once the Higgs boson mass $m_h \simeq 125\,\text{GeV}$ and vacuum expectation value $v \simeq 246\,\text{GeV}$ are fixed. Any deviation from these predictions would provide evidence for physics beyond the SM, such as extended scalar sectors, composite Higgs models, or mechanisms for EW baryogenesis. Reviews of such beyond-the-SM (BSM) scenarios can be found, for example, in Refs. [3, 4].

At the LHC, double-Higgs-boson production provides the primary direct probe of the Higgs boson trilinear self-coupling, while loop-induced effects in single-Higgs-boson processes yield complementary constraints [5–13]. These measurements are typically interpreted in the κ-framework [14–16], which parametrises deviations from the SM through

$$\kappa_3 = \frac{\lambda_3}{\lambda_3^{\rm SM}}\,, \tag{2}$$

assuming all other couplings take their SM values. Using the full $\sqrt{s} = 13\,\text{TeV}$ LHC Run 2 data set, the ATLAS and CMS collaborations obtain the combined constraint

$$-0.71 < \kappa_3 < 6.1 \quad (\text{ATLAS} + \text{CMS}) \tag{3}$$

at 95% confidence level (CL) [17]. The HL-LHC, with an integrated luminosity of $3\,\text{ab}^{-1}$ per experiment, corresponding to a combined ATLAS–CMS exposure of $6\,\text{ab}^{-1}$, is expected to improve this sensitivity to

$$0.5 < \kappa_3 < 1.7 \quad \text{(HL-LHC)}\,. \tag{4}$$

This projection is taken from Ref. [50].

Constraining the quartic Higgs boson self-coupling is considerably more challenging. It is conventionally parametrised as

$$\kappa_4 = \frac{\lambda_4}{\lambda_4^{\text{SM}}}\,, \tag{5}$$

and triple-Higgs-boson production provides its only direct probe at hadron colliders. However, the tiny SM cross section of approximately 100 ab at $\sqrt{s} = 14\,\text{TeV}$, calculated at next-to-next-to-leading order (NNLO) in QCD [18], renders such measurements experimentally very challenging. Current ATLAS [19] and CMS [20] searches yield

$$-230 < \kappa_4 < 240 \quad \text{(ATLAS)}\,, \qquad -190 < \kappa_4 < 190 \quad \text{(CMS)}\,, \tag{6}$$

at 95% CL, assuming all other couplings are SM-like, in particular $\kappa_3 = 1$. A recent HL-LHC study [21] projects these bounds to improve to

$$-81 < \kappa_4 < 89 \quad \text{(HL-LHC)} \tag{7}$$

for a combined ATLAS–CMS exposure of $6\,\text{ab}^{-1}$, corresponding to $3\,\text{ab}^{-1}$ per experiment. In addition to triple-Higgs-boson production, double-Higgs-boson production is also sensitive to κ_4 through two-loop EW corrections [21–24], although these constraints are generally weaker.

In this note, we present a comprehensive study of triple-Higgs-boson production within the κ-framework. We derive complementary compact parametrisations of the inclusive gluon-fusion cross section. The first retains the dependence on the Higgs boson trilinear and quartic self-couplings, the top-quark Yukawa coupling modifier κ_t, and the two-top–two-Higgs-boson interaction κ_{2t} for $\kappa_{3t} = 0$. The second incorporates the two-top–three-Higgs-boson interaction κ_{3t} for $\kappa_t = 1$ and $\kappa_{2t} = 0$. These expressions provide practical tools for interpreting current and future LHC measurements. Using them, we investigate the sensitivity of current LHC data and HL-LHC projections to the Higgs boson self-couplings and study the impact of correlated modifications of the top–Higgs-boson interactions. We further demonstrate that differential kinematic distributions retain significant sensitivity even in scenarios with nearly degenerate inclusive cross sections, highlighting the importance of shape information in future triple-Higgs-boson searches.

2 Triple-Higgs-boson production in the κ-framework

Triple-Higgs-boson production at hadron colliders has been the subject of extensive theoretical investigation over the past two decades [18, 21, 22, 25–41]. Within the SM, the gluon-fusion process $gg \to 3h$ is dominated by the one-loop topologies shown in the upper and middle rows of Figure 1. In the figure, green (blue) boxes denote insertions of the Higgs boson trilinear (quartic) self-coupling, while orange circles represent the top-quark Yukawa coupling. The two diagrams in the lower row illustrate representative BSM contributions involving a contact interaction between two top quarks and two Higgs bosons. In the SM, bottom-quark loops modify the $gg \to 3h$ production cross section by only about 0.2% at LHC energies [39]; they are nevertheless retained in the numerical calculations below.

The relevant terms in the κ-framework Lagrangian are

$$\mathcal{L}_\kappa \supset -\kappa_3 v \lambda h^3 - \kappa_4 \frac{\lambda}{4} h^4 - \kappa_t \frac{y_t}{\sqrt{2}}\,\bar{t}th - \kappa_{2t} \frac{y_t}{\sqrt{2}v}\,\bar{t}th^2 - \kappa_{3t} \frac{y_t}{2\sqrt{2}v^2}\,\bar{t}th^3\,, \tag{8}$$

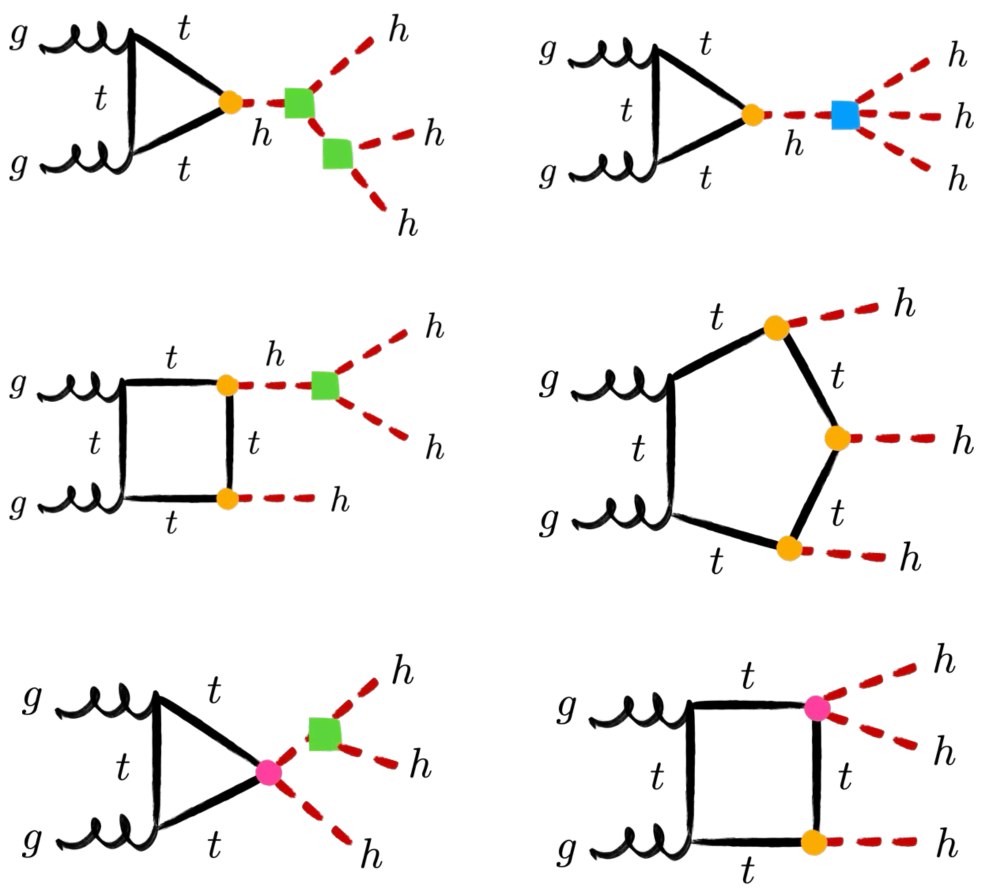

Figure 1: Representative one-loop Feynman diagrams contributing to triple-Higgs-boson production via gluon fusion, illustrating the triangle, box, and pentagon topologies. Green (blue) boxes denote the Higgs boson trilinear (quartic) self-coupling, while orange (magenta) circles indicate top-quark couplings involving one (two) Higgs bosons.

where κ_3 and κ_4 were introduced in Eqs. (2) and (5), respectively, and $y_t = \sqrt{2}m_t/v$, with m_t denoting the top-quark mass. The parameter κ_t thus encodes modifications of the top-quark Yukawa coupling, while κ_{2t} and κ_{3t} parametrise contact interactions between two top quarks and two or three Higgs bosons, respectively. The final term can equivalently be written as $-\kappa_{3t} m_t \bar{t}th^3/(2v^3)$ [35]. The SM is recovered from Eq. (8) for $\kappa_3 = \kappa_4 = \kappa_t = 1$ and $\kappa_{2t} = \kappa_{3t} = 0$.

2.1 Including the top-quark Yukawa and two-top–two-Higgs-boson interactions

To determine the dependence of the inclusive $gg \to 3h$ production cross section on κ_3, κ_4, κ_t, and κ_{2t}, we employ the leading-order (LO) matrix elements calculated in Ref. [39] and implemented in the `MCFM` [42–45] parton-level event generator. We use $m_h = 125\,\text{GeV}$ and $m_t = 173\,\text{GeV}$ as input parameters together with the `PDF4LHC21` parton distribution functions (PDFs) [46]. The renormalisation and factorisation scales, μ_R and μ_F, are chosen dynamically as $\mu_R = \mu_F = m_{3h}/2$, where m_{3h} denotes the invariant mass of the three-Higgs-boson system. At centre-of-mass energies $\sqrt{s} = 13\,\text{TeV}$, $13.6\,\text{TeV}$, and $14\,\text{TeV}$, and with $\kappa_{3t} = 0$, we

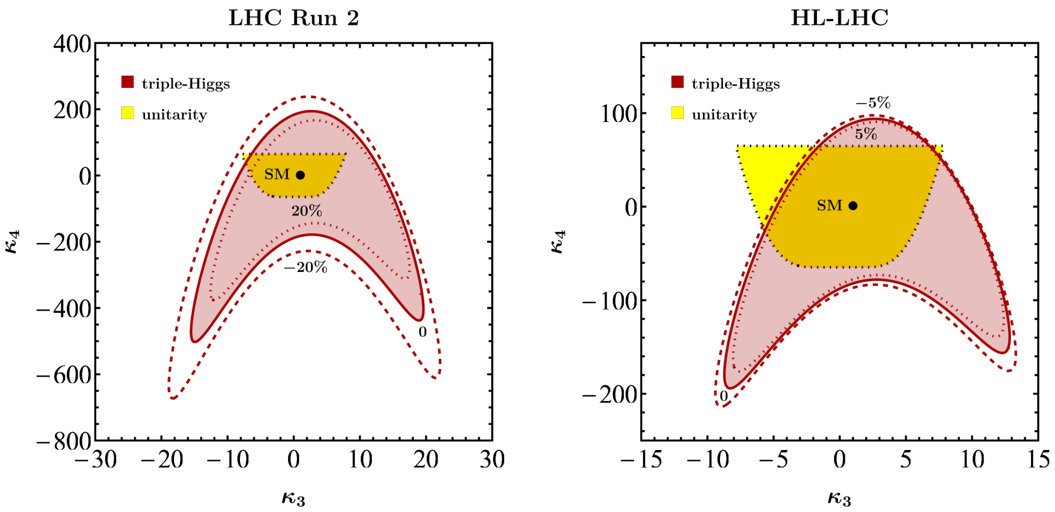

Figure 2: Constraints in the κ_3–κ_4 plane at LHC Run 2 (left) and the HL-LHC (right). The red contours indicate the allowed 95% CL regions obtained from triple-Higgs-boson production in gluon fusion for three representative values of κ_t, as specified by the contour labels. We assume $\kappa_{2t} = \kappa_{3t} = 0$. The SM prediction is denoted by black points. The yellow regions, enclosed by black dotted contours, represent the perturbative-unitarity bounds derived from tree-level $hh \to hh$ scattering.

parametrise the LO cross section as

$$\sigma_{\rm LO}^{\sqrt{s}} = \left(\sigma_{\rm LO}^{\sqrt{s}}\right)_{\rm SM} \sum_{n,m} c_{nm}^{\sqrt{s}}(\kappa_t, \kappa_{2t}) \left(\Delta\kappa_3\right)^n \left(\Delta\kappa_4\right)^m , \tag{9}$$

where $\Delta\kappa_i = \kappa_i - 1$ for $i = 3, 4$. The corresponding LO SM cross sections are

$$\sigma_{\rm LO}^{13\,\rm TeV} = 42.1\,\text{ab}\,, \qquad \sigma_{\rm LO}^{13.6\,\rm TeV} = 47.9\,\text{ab}\,, \qquad \sigma_{\rm LO}^{14\,\rm TeV} = 51.0\,\text{ab}\,. \tag{10}$$

For $\sqrt{s} = 13\,$TeV, the coefficients $c_{nm}^{\sqrt{s}}(\kappa_t, \kappa_{2t})$ appearing in Eq. (9) are

$$\begin{aligned} c_{00}^{13\,\rm TeV}(\kappa_t, \kappa_{2t}) = & -6.05 \times 10^{-4} + 0.0384\kappa_t^2 - 0.350\kappa_t^3 + 1.47\kappa_t^4 - 3.04\kappa_t^5 + 2.89\kappa_t^6 \\ & + \kappa_{2t}\left(1.61\kappa_t^2 - 7.82\kappa_t^3 + 15.6\kappa_t^4 - 14.8\kappa_t^5\right) \\ & + \kappa_{2t}^2\left(6.96 - 22.3\kappa_t + 37.0\kappa_t^2\right), \end{aligned} \tag{11}$$

$$c_{01}^{13\,\rm TeV}(\kappa_t, \kappa_{2t}) = 0.0829\kappa_t^2 - 0.264\kappa_t^3 + 0.0895\kappa_t^4 + \kappa_{2t}\left(0.673\kappa_t - 1.12\kappa_t^2\right), \tag{12}$$

$$c_{02}^{13\,\rm TeV}(\kappa_t, \kappa_{2t}) = 0.0169\kappa_t^2 , \tag{13}$$

$$\begin{aligned} c_{10}^{13\,\rm TeV}(\kappa_t, \kappa_{2t}) = & \, 0.211\kappa_t^2 - 1.45\kappa_t^3 + 3.65\kappa_t^4 - 3.28\kappa_t^5 \\ & + \kappa_{2t}\left(3.49\kappa_t - 13.4\kappa_t^2 + 15.6\kappa_t^3\right) + \kappa_{2t}^2\left(13.9 - 22.3\kappa_t\right), \end{aligned} \tag{14}$$

$$c_{11}^{13\,\rm TeV}(\kappa_t, \kappa_{2t}) = 0.0978\kappa_t^2 - 0.264\kappa_t^3 + 0.673\kappa_{2t}\kappa_t , \tag{15}$$

$$c_{20}^{13\,\rm TeV}(\kappa_t, \kappa_{2t}) = 0.288\kappa_t^2 - 1.25\kappa_t^3 + 1.84\kappa_t^4 + \kappa_{2t}\left(2.81\kappa_t - 6.70\kappa_t^2\right) + 6.96\kappa_{2t}^2 , \tag{16}$$

$$c_{21}^{13\,\rm TeV}(\kappa_t, \kappa_{2t}) = 0.0489\kappa_t^2 , \tag{17}$$

$$c_{30}^{13\,\rm TeV}(\kappa_t, \kappa_{2t}) = 0.159\kappa_t^2 - 0.417\kappa_t^3 + 0.938\kappa_{2t}\kappa_t , \tag{18}$$

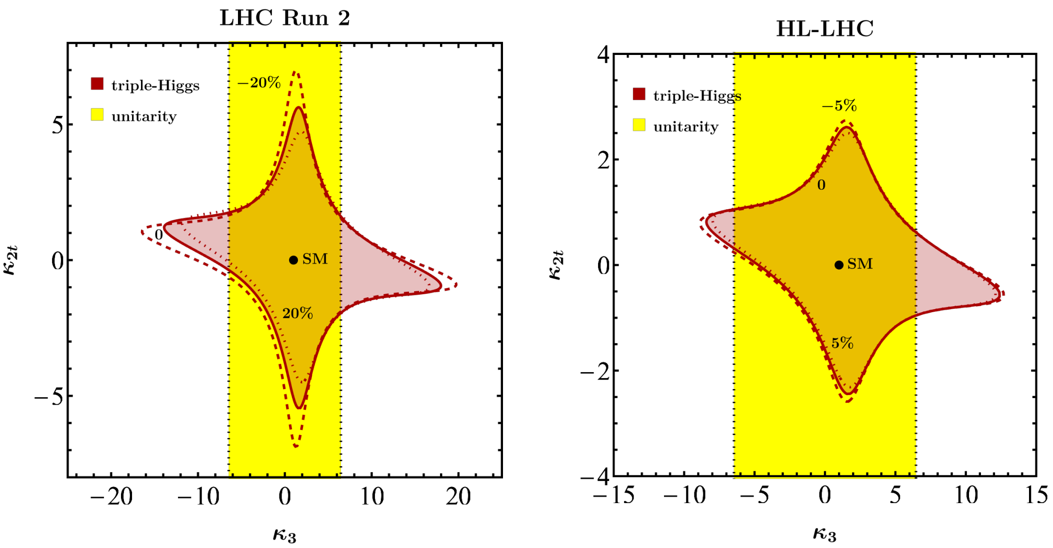

Figure 3: As in Figure 2, but in the κ_3–κ_{2t} plane at LHC Run 2 (left) and the HL-LHC (right). We fix $\kappa_4 = 1$ and $\kappa_{3t} = 0$, while κ_t takes the values indicated by the contour labels.

$$c_{40}^{13\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0396\kappa_t^2\,. \tag{19}$$

At $\sqrt{s} = 13.6\,\mathrm{TeV}$, the corresponding coefficients read

$$\begin{aligned} c_{00}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = &-9.34\times10^{-3} + 0.0572\kappa_t^2 - 0.446\kappa_t^3 + 1.68\kappa_t^4 - 3.17\kappa_t^5 + 2.89\kappa_t^6 \\ &+ \kappa_{2t}\left(1.59\kappa_t^2 - 7.69\kappa_t^3 + 15.3\kappa_t^4 - 14.6\kappa_t^5\right) \\ &+ \kappa_{2t}^2\left(6.89 - 22.1\kappa_t + 36.8\kappa_t^2\right), \end{aligned} \tag{20}$$

$$c_{01}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0814\kappa_t^2 - 0.260\kappa_t^3 + 0.0866\kappa_t^4 + \kappa_{2t}\left(0.665\kappa_t - 1.10\kappa_t^2\right), \tag{21}$$

$$c_{02}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0167\kappa_t^2\,, \tag{22}$$

$$\begin{aligned} c_{10}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = &\,0.204\kappa_t^2 - 1.40\kappa_t^3 + 3.56\kappa_t^4 - 3.21\kappa_t^5 \\ &+ \kappa_{2t}\left(3.43\kappa_t - 13.2\kappa_t^2 + 15.3\kappa_t^3\right) + \kappa_{2t}^2\left(13.8 - 22.1\kappa_t\right), \end{aligned} \tag{23}$$

$$c_{11}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0961\kappa_t^2 - 0.260\kappa_t^3 + 0.665\kappa_{2t}\kappa_t\,, \tag{24}$$

$$c_{20}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.283\kappa_t^2 - 1.23\kappa_t^3 + 1.81\kappa_t^4 + \kappa_{2t}\left(2.77\kappa_t - 6.59\kappa_t^2\right) + 6.89\kappa_{2t}^2\,, \tag{25}$$

$$c_{21}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0480\kappa_t^2\,, \tag{26}$$

$$c_{30}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.155\kappa_t^2 - 0.408\kappa_t^3 + 0.922\kappa_{2t}\kappa_t\,, \tag{27}$$

$$c_{40}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0388\kappa_t^2\,. \tag{28}$$

Finally, for $\sqrt{s} = 14\,\mathrm{TeV}$, we obtain

$$\begin{aligned} c_{00}^{14\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = &-2.18\times10^{-3} + 0.0323\kappa_t^2 - 0.311\kappa_t^3 + 1.35\kappa_t^4 - 2.92\kappa_t^5 + 2.84\kappa_t^6 \\ &+ \kappa_{2t}\left(1.61\kappa_t^2 - 7.79\kappa_t^3 + 15.5\kappa_t^4 - 14.7\kappa_t^5\right) \\ &+ \kappa_{2t}^2\left(6.99 - 22.4\kappa_t + 37.4\kappa_t^2\right), \end{aligned} \tag{29}$$

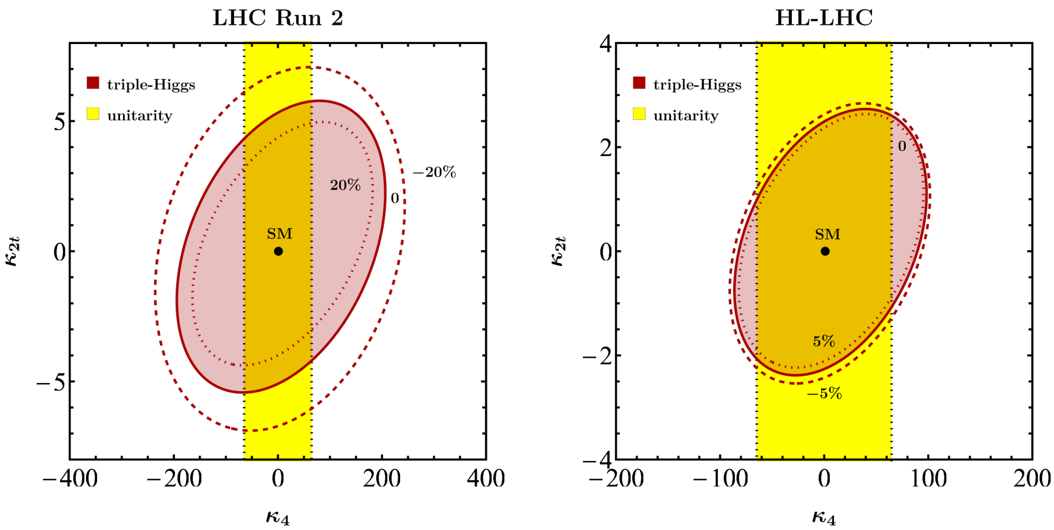

Figure 4: As in Figure 2, but in the κ_4–κ_{2t} plane at LHC Run 2 (left) and the HL-LHC (right). We fix $\kappa_3 = 1$ and $\kappa_{3t} = 0$, while κ_t takes the values indicated by the contour labels.

$$c_{01}^{14\,\text{TeV}}(\kappa_t,\kappa_{2t}) = 0.0822\kappa_t^2 - 0.262\kappa_t^3 + 0.0865\kappa_t^4 + \kappa_{2t}\left(0.672\kappa_t - 1.12\kappa_t^2\right), \tag{30}$$

$$c_{02}^{14\,\text{TeV}}(\kappa_t,\kappa_{2t}) = 0.0169\kappa_t^2, \tag{31}$$

$$c_{10}^{14\,\text{TeV}}(\kappa_t,\kappa_{2t}) = 0.194\kappa_t^2 - 1.39\kappa_t^3 + 3.58\kappa_t^4 - 3.24\kappa_t^5 + \kappa_{2t}\left(3.46\kappa_t - 13.3\kappa_t^2 + 15.5\kappa_t^3\right) + \kappa_{2t}^2(14.0 - 22.4\kappa_t), \tag{32}$$

$$c_{11}^{14\,\text{TeV}}(\kappa_t,\kappa_{2t}) = 0.0968\kappa_t^2 - 0.262\kappa_t^3 + 0.672\kappa_{2t}\kappa_t, \tag{33}$$

$$c_{20}^{14\,\text{TeV}}(\kappa_t,\kappa_{2t}) = 0.284\kappa_t^2 - 1.23\kappa_t^3 + 1.82\kappa_t^4 + \kappa_{2t}\left(2.79\kappa_t - 6.65\kappa_t^2\right) + 6.99\kappa_{2t}^2, \tag{34}$$

$$c_{21}^{14\,\text{TeV}}(\kappa_t,\kappa_{2t}) = 0.0484\kappa_t^2, \tag{35}$$

$$c_{30}^{14\,\text{TeV}}(\kappa_t,\kappa_{2t}) = 0.156\kappa_t^2 - 0.411\kappa_t^3 + 0.931\kappa_{2t}\kappa_t, \tag{36}$$

$$c_{40}^{14\,\text{TeV}}(\kappa_t,\kappa_{2t}) = 0.0391\kappa_t^2. \tag{37}$$

We note that the coefficients $c_{nm}^{\sqrt{s}}(\kappa_t,\kappa_{2t})$ depend only mildly on the collision energy over the range considered. Likewise, the choice of PDFs is expected to have a negligible numerical impact on the parametrisations given in Eqs. (11)–(37). Higher-order QCD corrections to the coefficients $c_{nm}^{\sqrt{s}}(\kappa_t,\kappa_{2t})$ are currently unknown, although they could, in principle, be estimated in the infinite-top-quark-mass limit. In the SM, these corrections increase the inclusive cross section by a K-factor of approximately 1.7 at NNLO in QCD [18]. Parametrisations of the triple-Higgs-boson gluon-fusion cross section at the LHC similar to Eqs. (9)–(37) have also been presented in Refs. [35, 36, 38, 39].

The two panels in Figure 2 illustrate the sensitivity to the Higgs boson self-couplings κ_3 and κ_4 using the full LHC Run 2 data set and HL-LHC projections corresponding to $3\,\text{ab}^{-1}$ per experiment. The red contours indicate the 95% CL constraints derived from triple-Higgs-boson production in gluon fusion using the parametrisations in Eqs. (11)–(19) and Eqs. (29)–(37), with the normalisation to the corresponding LO SM cross section given in Eq. (10). For LHC Run 2, the triple-Higgs-boson constraints are obtained by imposing $\mu_{3h}^{\text{LHC Run 2}} < 588$ [20], while for the HL-LHC we assume a projected upper limit of $\mu_{3h}^{\text{HL-LHC}} < 125$ [47]. Throughout this analysis, we set $\kappa_{2t} = \kappa_{3t} = 0$, and the black points indicate the SM prediction. The left panel shows that shifts of $\Delta\kappa_t = \pm 20\%$, where $\Delta\kappa_t = \kappa_t - 1$, can significantly affect the resulting constraints. Such shifts remain allowed at 95% CL by LHC Run 2 data [48, 49]. This behaviour was already pointed out in Ref. [41] and originates from the strong power-law dependence of the coefficients $c_{nm}^{\sqrt{s}}(\kappa_t, \kappa_{2t})$ appearing in Eqs. (11)–(37) on κ_t. By the end of the HL-LHC, the top-quark Yukawa coupling is expected to be measured with a precision of $\Delta\kappa_t = \pm 5\%$ at 95% CL [50]. Consequently, its impact on the $gg \to 3h$ constraints in the κ_3–κ_4 plane will be substantially reduced, as illustrated in the right panel. Finally, we note that the projected HL-LHC constraints from $gg \to 3h$ are comparable to those imposed by perturbative unitarity [34, 51, 52], shown as the yellow region enclosed by the black-dotted contour in both panels.

The panels in Figures 3 and 4 show the expected sensitivity in the κ_3–κ_{2t} and κ_4–κ_{2t} planes, respectively. The red contours represent the 95% CL constraints from triple-Higgs-boson production in gluon fusion, obtained following the procedure described above. Apart from κ_t, which varies as indicated by the contour labels, all parameters not shown on the axes are fixed to their SM values, and the black points indicate the SM prediction. As in Figure 2, the left panels demonstrate that variations of the top-quark Yukawa coupling of $\Delta\kappa_t = \pm 20\%$ can have a significant impact on the resulting constraints, whereas the smaller variations $\Delta\kappa_t = \pm 5\%$ considered in the right panels lead to only minor changes. For κ_{2t}, triple-Higgs-boson production at LHC Run 2 yields $-5.1 < \kappa_{2t} < 5.3$, while the HL-LHC is expected to improve this range to $-2.3 < \kappa_{2t} < 2.5$ at 95% CL. These limits are obtained by fixing $\kappa_3 = \kappa_4 = \kappa_t = 1$ and $\kappa_{3t} = 0$. We note that, under the same assumptions, the 95% CL constraint on the double-Higgs-boson production signal strength obtained at LHC Run 2, $\mu_{2h}^{\text{LHC Run 2}} < 2.5$ [17], already implies $-0.19 < \kappa_{2t} < 0.72$. The quoted bound is obtained using the expressions given in Refs. [53, 54]. Consequently, triple-Higgs-boson production is not expected to provide a stronger constraint on κ_{2t} than double-Higgs-boson production in a single-parameter analysis. It can, however, provide complementary information in multi-parameter fits, in particular when these involve κ_4, which can only be weakly constrained by double-Higgs-boson production [21–24].

2.2 Including the two-top–three-Higgs-boson interaction

We next extend the inclusive-rate parametrisation to the contact interaction between two top quarks and three Higgs bosons in Eq. (8). We generate loop-induced LO parton-level samples with `MadGraph5_aMC@NLO` [55], using the same Higgs boson mass m_h, top-quark mass m_t, PDFs, and scale choice as above. In this scan, $\kappa_t = 1$ and $\kappa_{2t} = 0$ are fixed.

Following Eq. (9), we write the extended cross section as

$$\sigma_{\text{LO}}^{\sqrt{s}} = \left(\sigma_{\text{LO}}^{\sqrt{s}}\right)_{\text{SM}} \sum_{n,m} c_{nm}^{\sqrt{s}}(\kappa_{3t}) \left(\Delta\kappa_3\right)^n \left(\Delta\kappa_4\right)^m . \tag{38}$$

This is an exact regrouping of the LO coupling basis at the fixed values $\kappa_t = 1$ and $\kappa_{2t} = 0$. Since the amplitude is linear in κ_{3t}, c_{00} is at most quadratic in κ_{3t}, while c_{01}, c_{10}, and c_{20} are at most linear; the remaining nonzero coefficient functions are constant. The fitted SM cross sections are

$$\left(\sigma_{\text{LO}}^{13\,\text{TeV}}\right)_{\text{SM}} = 42.12\,\text{ab}\,, \qquad \left(\sigma_{\text{LO}}^{13.6\,\text{TeV}}\right)_{\text{SM}} = 47.40\,\text{ab}\,, \qquad \left(\sigma_{\text{LO}}^{14\,\text{TeV}}\right)_{\text{SM}} = 51.08\,\text{ab}\,. \tag{39}$$

They agree to within 1.1% with the independent values in Eq. (10).

The normalisation fixes $c_{00}^{\sqrt{s}}(0) = 1$. The fitted coefficient functions are listed below. We retain more significant figures than in the first parametrisation to preserve the numerical accuracy of the fit.

For $\sqrt{s} = 13\,\mathrm{TeV}$, the non-vanishing coefficient functions are

$$c_{00}^{13\,\mathrm{TeV}}(\kappa_{3t}) = 1 - 2.83183\,\kappa_{3t} + 15.81829\,\kappa_{3t}^2\,, \tag{40}$$
$$c_{01}^{13\,\mathrm{TeV}}(\kappa_{3t}) = -0.0914560 + 0.793591\,\kappa_{3t}\,, \tag{41}$$
$$c_{02}^{13\,\mathrm{TeV}}(\kappa_{3t}) = 0.0168979\,, \tag{42}$$
$$c_{10}^{13\,\mathrm{TeV}}(\kappa_{3t}) = -0.865038 - 3.92841\,\kappa_{3t}\,, \tag{43}$$
$$c_{11}^{13\,\mathrm{TeV}}(\kappa_{3t}) = -0.167158\,, \tag{44}$$
$$c_{20}^{13\,\mathrm{TeV}}(\kappa_{3t}) = 0.872294 + 0.900894\,\kappa_{3t}\,, \tag{45}$$
$$c_{21}^{13\,\mathrm{TeV}}(\kappa_{3t}) = 0.0487935\,, \tag{46}$$
$$c_{30}^{13\,\mathrm{TeV}}(\kappa_{3t}) = -0.259375\,, \tag{47}$$
$$c_{40}^{13\,\mathrm{TeV}}(\kappa_{3t}) = 0.0395354\,. \tag{48}$$

For $\sqrt{s} = 13.6\,\mathrm{TeV}$, they are

$$c_{00}^{13.6\,\mathrm{TeV}}(\kappa_{3t}) = 1 - 2.87872\,\kappa_{3t} + 16.21306\,\kappa_{3t}^2\,, \tag{49}$$
$$c_{01}^{13.6\,\mathrm{TeV}}(\kappa_{3t}) = -0.0909154 + 0.800038\,\kappa_{3t}\,, \tag{50}$$
$$c_{02}^{13.6\,\mathrm{TeV}}(\kappa_{3t}) = 0.0168638\,, \tag{51}$$
$$c_{10}^{13.6\,\mathrm{TeV}}(\kappa_{3t}) = -0.866564 - 3.95165\,\kappa_{3t}\,, \tag{52}$$
$$c_{11}^{13.6\,\mathrm{TeV}}(\kappa_{3t}) = -0.166501\,, \tag{53}$$
$$c_{20}^{13.6\,\mathrm{TeV}}(\kappa_{3t}) = 0.871502 + 0.897150\,\kappa_{3t}\,, \tag{54}$$
$$c_{21}^{13.6\,\mathrm{TeV}}(\kappa_{3t}) = 0.0485042\,, \tag{55}$$
$$c_{30}^{13.6\,\mathrm{TeV}}(\kappa_{3t}) = -0.257216\,, \tag{56}$$
$$c_{40}^{13.6\,\mathrm{TeV}}(\kappa_{3t}) = 0.0390715\,. \tag{57}$$

For $\sqrt{s} = 14\,\mathrm{TeV}$, they are

$$c_{00}^{14\,\mathrm{TeV}}(\kappa_{3t}) = 1 - 2.95187\,\kappa_{3t} + 16.47338\,\kappa_{3t}^2\,, \tag{58}$$
$$c_{01}^{14\,\mathrm{TeV}}(\kappa_{3t}) = -0.0921387 + 0.803365\,\kappa_{3t}\,, \tag{59}$$
$$c_{02}^{14\,\mathrm{TeV}}(\kappa_{3t}) = 0.0168553\,, \tag{60}$$
$$c_{10}^{14\,\mathrm{TeV}}(\kappa_{3t}) = -0.853992 - 3.93891\,\kappa_{3t}\,, \tag{61}$$
$$c_{11}^{14\,\mathrm{TeV}}(\kappa_{3t}) = -0.166145\,, \tag{62}$$
$$c_{20}^{14\,\mathrm{TeV}}(\kappa_{3t}) = 0.863440 + 0.899872\,\kappa_{3t}\,, \tag{63}$$
$$c_{21}^{14\,\mathrm{TeV}}(\kappa_{3t}) = 0.0482948\,, \tag{64}$$
$$c_{30}^{14\,\mathrm{TeV}}(\kappa_{3t}) = -0.255934\,, \tag{65}$$
$$c_{40}^{14\,\mathrm{TeV}}(\kappa_{3t}) = 0.0390078\,. \tag{66}$$

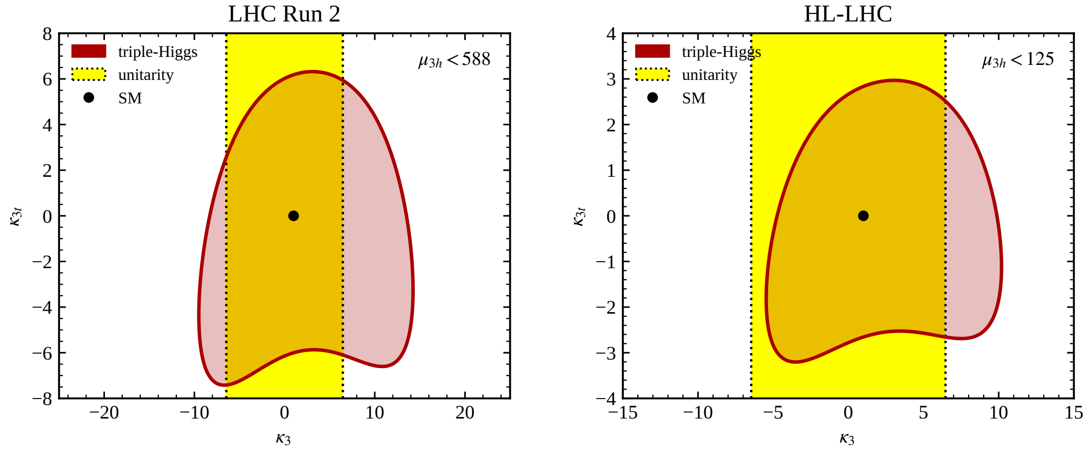

Figure 5: Constraints in the κ_3–κ_{3t} plane at LHC Run 2 (left) and the HL-LHC (right), obtained from Eq. (38) using $\mu_{3h} < 588$ and $\mu_{3h} < 125$, respectively. The red regions are allowed by the inclusive triple-Higgs-boson rate, the black points denote the SM, and the yellow vertical bands show the projection of the tree-level $hh \to hh$ perturbative-unitarity constraint. We fix $\kappa_4 = \kappa_t = 1$ and $\kappa_{2t} = 0$. The yellow band constrains κ_3 only; no independent unitarity condition on κ_{3t} is imposed.

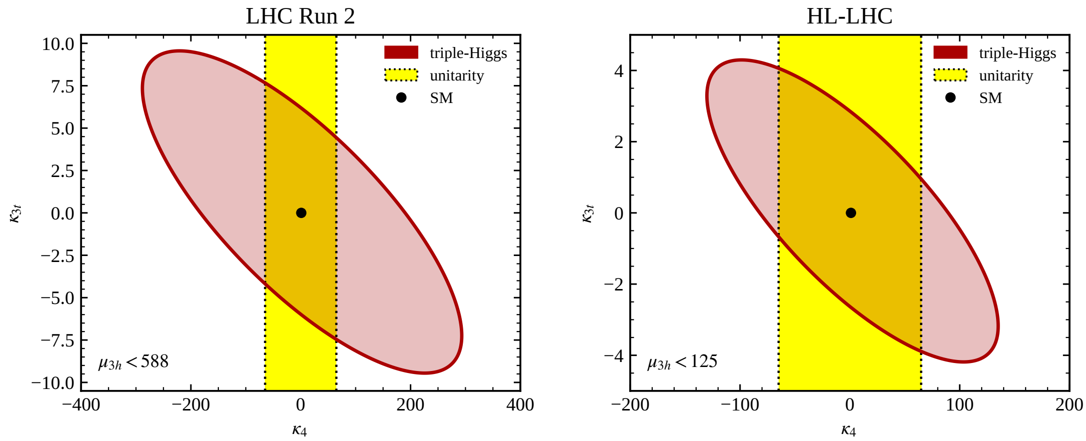

Figure 6: As in Figure 5, but in the κ_4–κ_{3t} plane with $\kappa_3 = \kappa_t = 1$ and $\kappa_{2t} = 0$. The yellow band is the projection of the $hh \to hh$ perturbative-unitarity condition onto κ_4.

Figures 5 and 6 are the κ_{3t} analogues of Figures 3 and 4. Fixing all other couplings to their SM values yields

$$-6.00 < \kappa_{3t} < 6.18 \qquad (\text{LHC Run 2}), \qquad -2.66 < \kappa_{3t} < 2.83 \qquad (\text{HL-LHC}). \tag{67}$$

The quadratic form also exposes an inclusive-rate degeneracy. At $\kappa_3 = \kappa_4 = 1$, the two solutions of $c_{00}^{\sqrt{s}}(\kappa_{3t}) = 1$ are $\kappa_{3t} = 0$ and an energy-dependent second solution near $\kappa_{3t} = 0.179$. Differential information is therefore essential for distinguishing these two solutions.

Modifications of the Higgs boson trilinear and quartic self-couplings affect both the inclusive $gg \to 3h$ cross section and the kinematic properties of the final state. Figure 7 compares normalised distributions for the SM with two BSM benchmarks. Both benchmarks give $\mu_{3h} \simeq 65$, and their inclusive cross sections differ by less than 5%, while their values of κ_3 and κ_4 satisfy the perturbative-unitarity constraints from tree-level $hh \to hh$ scattering. We

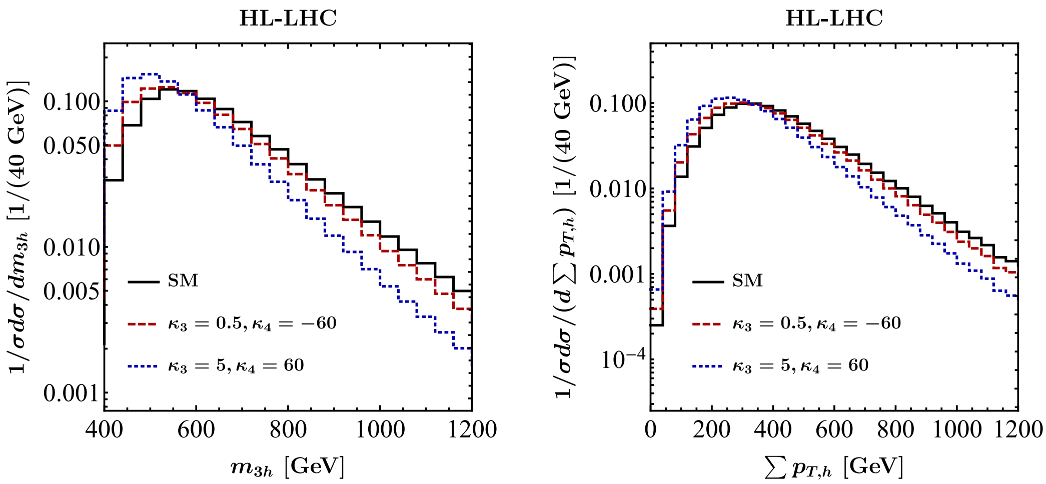

Figure 7: Normalised kinematic distributions for $gg \to 3h$ production at the HL-LHC in the SM (solid black line) and in two benchmark scenarios defined by κ_3 and κ_4 (red dashed and blue dotted lines). The left and right panels show the invariant mass of the three-Higgs-boson system, m_{3h}, and the scalar sum of the transverse momenta of the Higgs bosons, $\sum p_{T,h}$, respectively. All results are obtained for $\kappa_t = 1$ and $\kappa_{2t} = \kappa_{3t} = 0$.

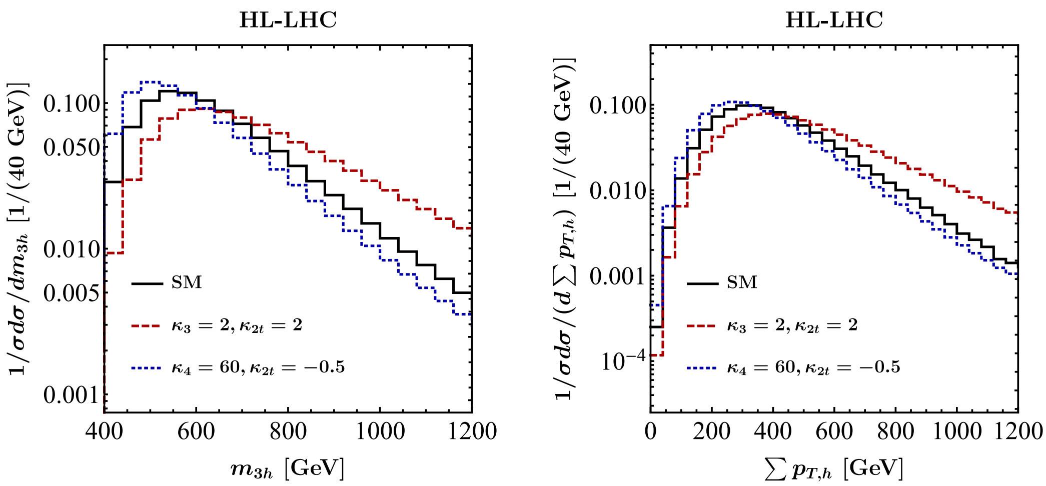

Figure 8: As in Figure 7, but for two further benchmark scenarios in the κ-framework, as indicated in the panels. The remaining parameters are fixed to their SM values, including $\kappa_{3t} = 0$.

set $\kappa_t = 1$ and $\kappa_{2t} = \kappa_{3t} = 0$. Variations of κ_t predominantly rescale the unnormalised spectra and therefore have little effect on their normalised shapes [41]. Nevertheless, the two benchmarks produce deviations of up to about 50% in the peak regions and high-energy tails of both distributions. Differential information can therefore strengthen future HL-LHC constraints beyond the projected interval in Eq. (7), which is obtained from the inclusive rate alone [21].

Figure 8 presents further normalised distributions for the SM and two BSM benchmarks in the κ-framework. Both benchmarks give $\mu_{3h} \simeq 75$, and their inclusive cross sections differ by less than 5%. The non-SM values of κ_3 or κ_4 satisfy the relevant perturbative-unitarity constraints. The values of κ_{2t} are indicated in the panels, and all parameters not specified by each benchmark are fixed to their SM values, including $\kappa_{3t} = 0$. As in Figure 7, sizeable shape distortions remain despite the near-degeneracy of the inclusive rates. This illustrates how differential information can help to resolve parameter degeneracies in future $gg \to 3h$ searches at the HL-LHC.

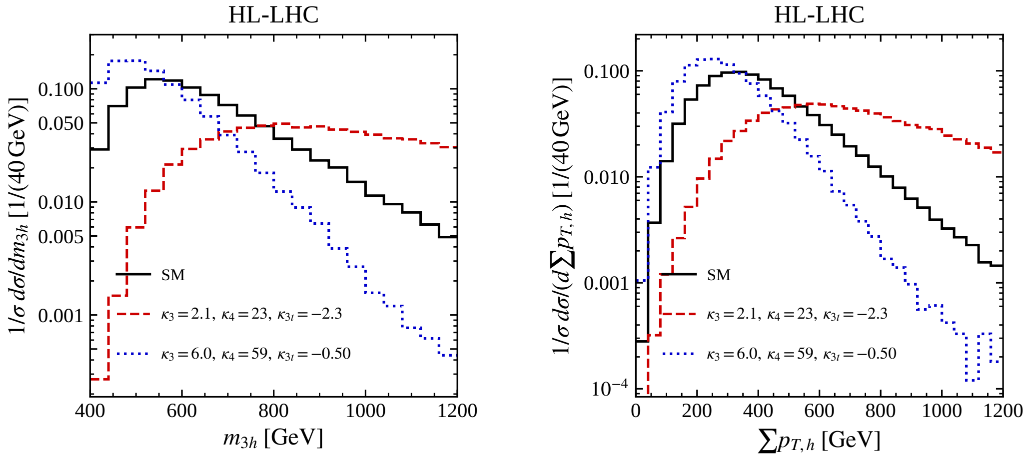

Figure 9: Normalised m_{3h} and $\sum p_{T,h}$ distributions at the HL-LHC for the SM and the two $\mu_{3h} \simeq 65$ benchmarks $(\kappa_3, \kappa_4, \kappa_{3t}) = (2.1, 23, -2.3)$ and $(6.0, 59, -0.50)$. The distributions use the same 40 GeV binning and plotting ranges as Figures 7 and 8. We fix $\kappa_t = 1$ and $\kappa_{2t} = 0$.

The two mixed-coupling benchmarks in Figure 9 provide a particularly clear example of enhanced but nearly degenerate inclusive rates, with $\mu_{3h} \approx 65$. Nevertheless, their normalised distributions differ substantially from the SM distributions and from one another. The $(2.1, 23, -2.3)$ benchmark shifts weight towards larger invariant masses and transverse momenta, whereas the $(6.0, 59, -0.50)$ benchmark enhances the near-threshold and low-p_T regions.

3 Conclusions

Unlike the Higgs boson couplings to EW gauge bosons and third-generation fermions, which are already being measured with increasing precision, the Higgs boson self-interactions remain largely unconstrained by LHC Run 2 data. This leaves considerable room for BSM scenarios that modify the shape of the Higgs potential. Indeed, both theoretical considerations and explicit ultraviolet-complete constructions demonstrate that deviations of the Higgs boson self-couplings exceeding 200% can be consistent with existing single-Higgs-boson measurements without requiring fine-tuned parameter choices [4]. Measurements of multi-Higgs-boson production at the HL-LHC and future colliders therefore provide a unique opportunity to probe largely unexplored regions of parameter space and offer significant discovery potential, even in scenarios where single-Higgs-boson observables remain close to their SM predictions.

In this note, we have derived compact 'pocket formulae' for triple-Higgs-boson production within the κ-framework, enabling constraints on the Higgs boson trilinear and quartic self-couplings to be obtained from current and future measurements of the $gg \to 3h$ signal strength. The parametrisation in Eq. (9) retains the full dependence on κ_3, κ_4, κ_t, and κ_{2t} at LO in perturbation theory for $\kappa_{3t} = 0$. We have supplemented it with Eq. (38), which includes the complete LO dependence on κ_{3t} for $\kappa_t = 1$ and $\kappa_{2t} = 0$ at 13, 13.6, and 14 TeV. Since the derivation relies on flexible matrix-element implementations, further extensions to incorporate additional coupling modifications — such as those arising at next-to-leading order in Higgs effective field theory [54] — are comparatively straightforward.

Using these parametrisations, we have studied how current LHC data and HL-LHC projections constrain κ_3 and κ_4. We find that the presently allowed uncertainty in κ_t can have a substantial impact on the constraints in the κ_3–κ_4 plane. This dependence is significantly reduced at the HL-LHC, where the expected percent-level precision of the top-quark Yukawa coupling measurement will strongly restrict the corresponding parameter space. For κ_{2t}, which parametrises the interaction between two top quarks and two Higgs bosons, we find that triple-Higgs-boson production is unlikely to provide stronger bounds than double-Higgs-boson production in a single-parameter analysis. Nevertheless, it can yield complementary information in global multi-parameter fits, especially once κ_4 is included, as this coupling remains only weakly constrained by double-Higgs-boson measurements [21–24]. In a single-parameter interpretation, the current and projected upper limits on the triple-Higgs-boson rate give $-6.00 < \kappa_{3t} < 6.18$ and $-2.66 < \kappa_{3t} < 2.83$, respectively. The energy-dependent nonzero solution near $\kappa_{3t} = 0.179$ is rate-degenerate with the SM. We further identify mixed $(\kappa_3, \kappa_4, \kappa_{3t})$ benchmarks with nearly identical inclusive rates, $\mu_{3h} \simeq 65$, whose normalised kinematic distributions differ substantially from the SM distributions and from one another. These examples illustrate how differential information can help to lift inclusive-rate degeneracies.

The projected HL-LHC sensitivity to κ_4 in a single-parameter analysis remains modest, with constraints reaching only $|\kappa_4| \lesssim 80$, comparable to those derived from perturbative-unitarity arguments [34, 51, 52]. The proposed hadron-collider stage of the Future Circular Collider could improve the sensitivity to the Higgs boson quartic self-coupling by approximately one order of magnitude [22, 28–30, 32, 33, 35–38, 40]. Reaching a precision at the level of 50%, however, would likely require the capabilities of a future muon collider [56].

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